



科技行业 - AI长周期能见度提升

June 2025

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投资结论

观点、结论

- 关税+地缘政治+美国经济疑虑引发前期波动，后期反弹亦反映担忧趋缓加上基本面强劲，宏观不确定性仍在

半导体受AI驱动，然周期进入高原期，保守看SPE

- 台积电重申指引，并乐观看行业前景，然周期进入高原期，先进制程续强，成熟制程疲软
- 半导体WFE明年或持平到衰退，成熟制程放缓，N2或放缓。然中国国内设备为增长亮点

存储周期涨价优于预期，近存计算为新兴题材，看好利基型

- NAND、DRAM於2季度起涨，然涨幅延续至3季度，不确定性落在4Q25
- 高通8 Gen 6+或搭3D DRAM，手机端近存DRAM市场规模达\$1-1.5bn，疊加DDR4涨价，看好利基型

AI CSP资本支出较预期佳，上修ASIC，並看好GPGPU主线

- CSP资本支出较预期佳，Token爆量显示推理需求，生产面GB200改善
- ASIC预期於近月上修 (博通、MRVL、CLS、光模块、CCL)，关注CoWoS/WMCM，并持续看好NVDA、AMD

下游手机产业链需求仍淡

- iPhone 生产量提前导致库存升高，Apple Intelligence仍不明朗，Android手机平平

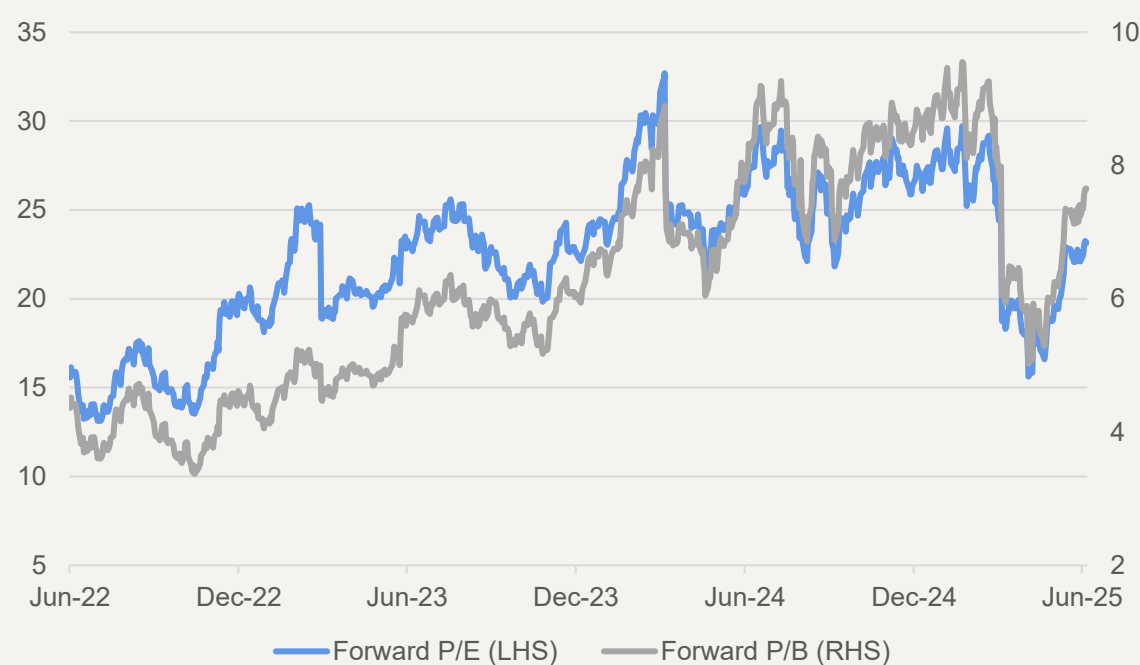
主要科技指数于关税战趋缓后大幅反弹

- 关税政策自4月下旬软化，过去2个月指数已低点大幅反弹，后续中美谈判或仍有不确定性，加上潜在半导体关税
- 个股的股价相对位置，如博通6月5日业绩强劲，但股价高位回调

费半指数P/E、P/B



MSCI Semi P/E、P/B



对等关税暂免，半导体调查仍在进行

对等关税暂受豁免

日期	关税政策
2月4日-3月4日	芬太尼关税：中20%，加、墨25%
4月5日	10%对等关税
4月8日	10%-125%对等关税最高税率 (中国适用125%)
4月9日	对等关税最高税率暂停90日 (中国除外) 至7月10日
5月14日	中国对等关税最高税率暂停90日 (下调至10%) 至8月14日
5月31日	301关税豁免期延长至8月31日

电子产品税率待定

产品	关税政策
消费电子产品	- 豁免对等关税 - 中国产品适用20%芬太尼税率 - 特朗普23日称非美国生产iPhone可能导致苹果产品面临25%税率
半导体	- 232关税政策调查中 - 美国商务部就半导体关税征求意见，于5月7日结束，媒体报导最早可能6月底征收关税

AI 政策限制

拜登政府时代出口管制框架废除

Bureau of Industry & Security
Office of Congressional and Public Affairs



FOR IMMEDIATE RELEASE | May 13, 2025 | Media Contact: OCPA@bis.doc.gov

Department of Commerce Rescinds Biden-Era Artificial Intelligence Diffusion Rule, Strengthens Chip-Related Export Controls

H20 -> 6000D / B40 ?

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
WASHINGTON, DC 20549

FORM 8-K

CURRENT REPORT
PURSUANT TO SECTION 13 OR 15(d) OF
THE SECURITIES EXCHANGE ACT OF 1934

Date of Report (Date of earliest event reported): April 9, 2025

NVIDIA CORPORATION
(Exact name of registrant as specified in its charter)

Delaware
(State or other jurisdiction
of incorporation)

0-23985
(Commission
File Number)

94-31775-49
(IRS Employer
Identification No.)

2788 San Tomas Expressway, Santa Clara, CA 95051
(Address of principal executive offices) (Zip Code)

Registrant's telephone number, including area code: (408) 486-2000

Not Applicable
(Former name or former address, if changed since last report)

Check the appropriate box below if the Form 8-K filing is intended to simultaneously satisfy the filing obligation of the registrant under any of the following provisions:

☐ Written communications pursuant to Rule 425 under the Securities Act (17 CFR 230.425)

☐ Soliciting material pursuant to Rule 144a-12 under the Exchange Act (17 CFR 240.144a-12)

☐ Pre-commencement communications pursuant to Rule 14d-2(b) under the Exchange Act (17 CFR 240.14d-2(b))

☐ Pre-commencement communications pursuant to Rule 13e-4(c) under the Exchange Act (17 CFR 240.13e-4(c))

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, \$0.001 per value per share	NVDA	The Nasdaq Global Select Market

Indicate by check mark whether the registrant is an emerging growth company as defined in Rule 405 of the Securities Act of 1933 (§230.405 of this chapter) or Rule 12b-2 of the Securities Exchange Act of 1934 (§240.12b-2 of this chapter).

Emerging Growth Company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Item 8.01 Other Events.

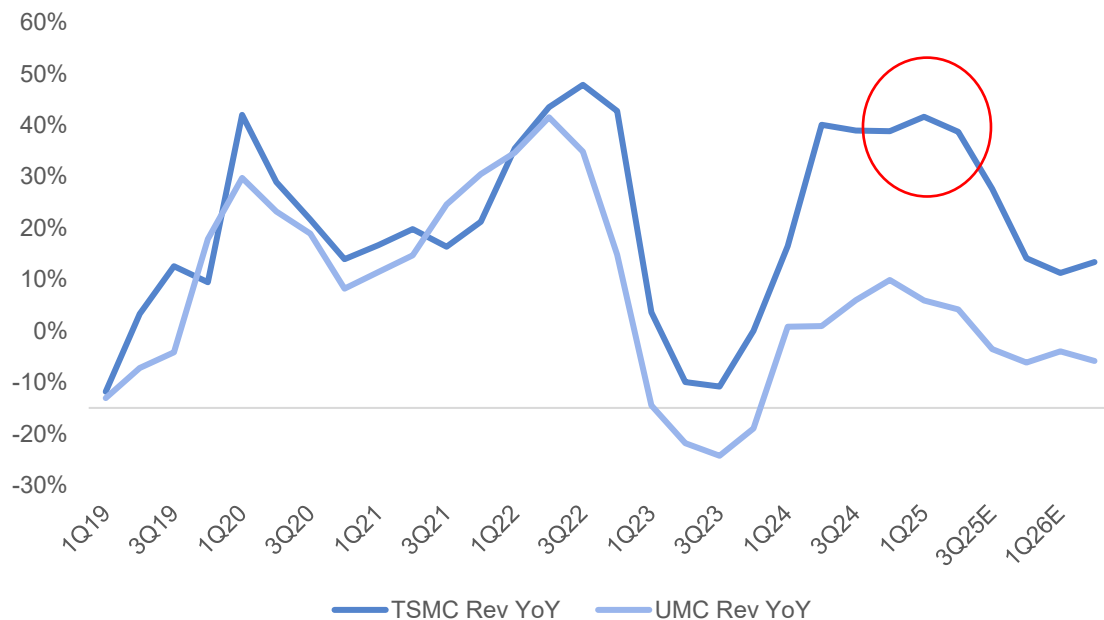
On April 9, 2025, the U.S. government, or USG, informed NVIDIA Corporation, or the Company, that the USG requires a license for export to China (including Hong Kong and Macau) and D-5 countries, or to companies headquartered or with an ultimate parent therein, of the Company's H20 integrated circuits and any other circuits achieving the H20's memory bandwidth, interconnect bandwidth, or combination thereof. The USG indicated that the license requirement addresses the risk that the covered products may be used in, or diverted to, a supercomputer in China. On April 14, 2025, the USG informed the Company that the license requirement will be in effect for the indefinite future.

The Company's first quarter of fiscal year 2026 ends on April 27, 2025. First quarter results are expected to include up to approximately \$5.5 billion of charges associated with H20 products for inventory, purchase commitments, and related reserves.

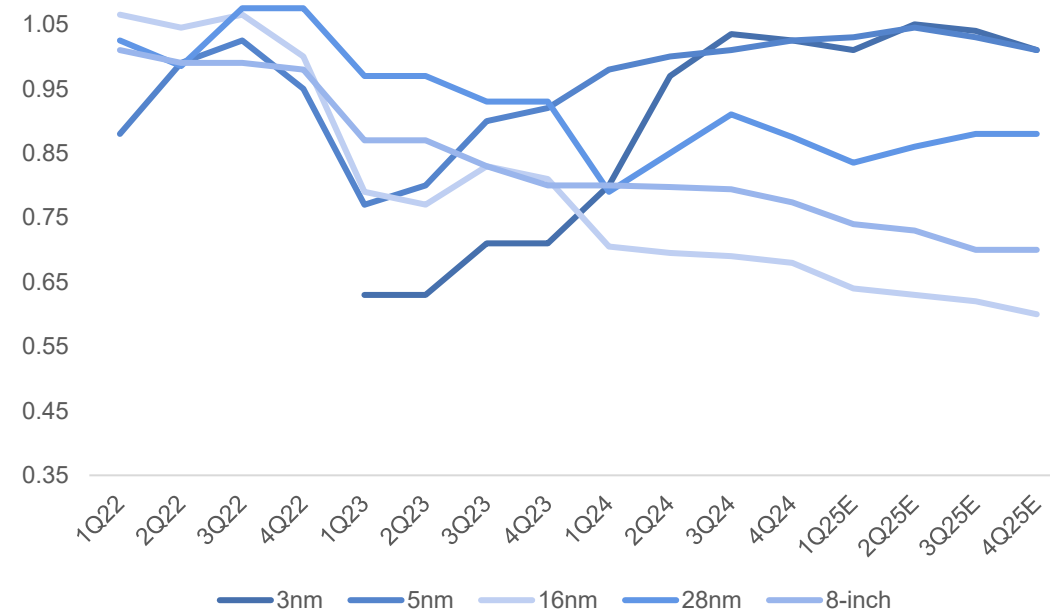
先进节点维持高位，周期进入高原期

- 预计台积电2025年1Q-3Q N3/N5 UTR 维持100%+, 然其他制程较弱
- 关注上半年是否提前预支需求

台积电 vs. 联电 收入同比

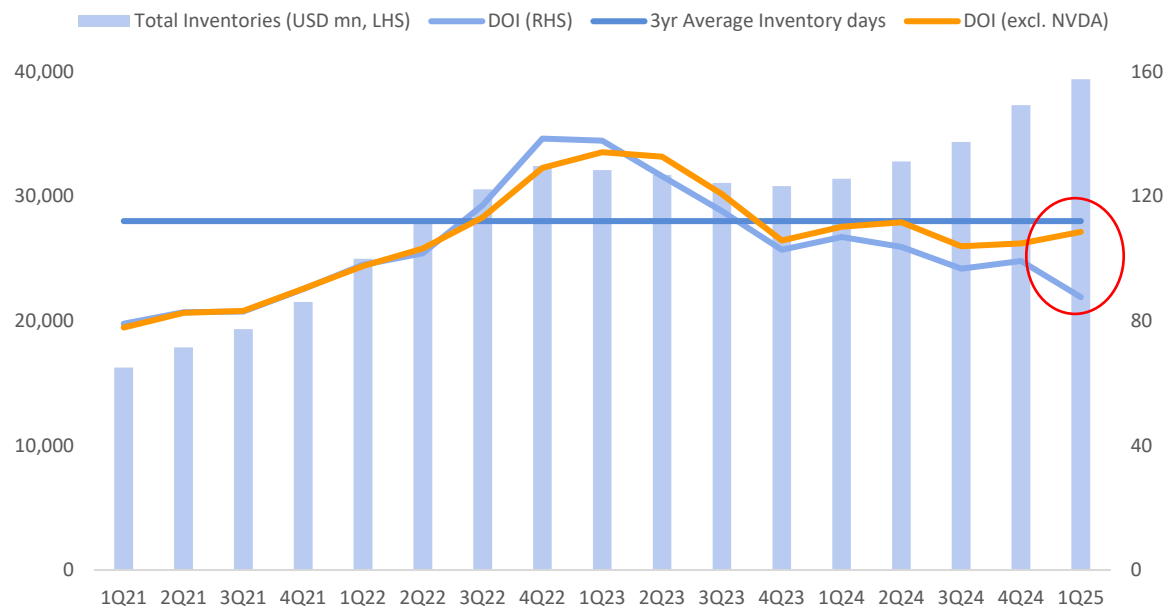


台积电UTR



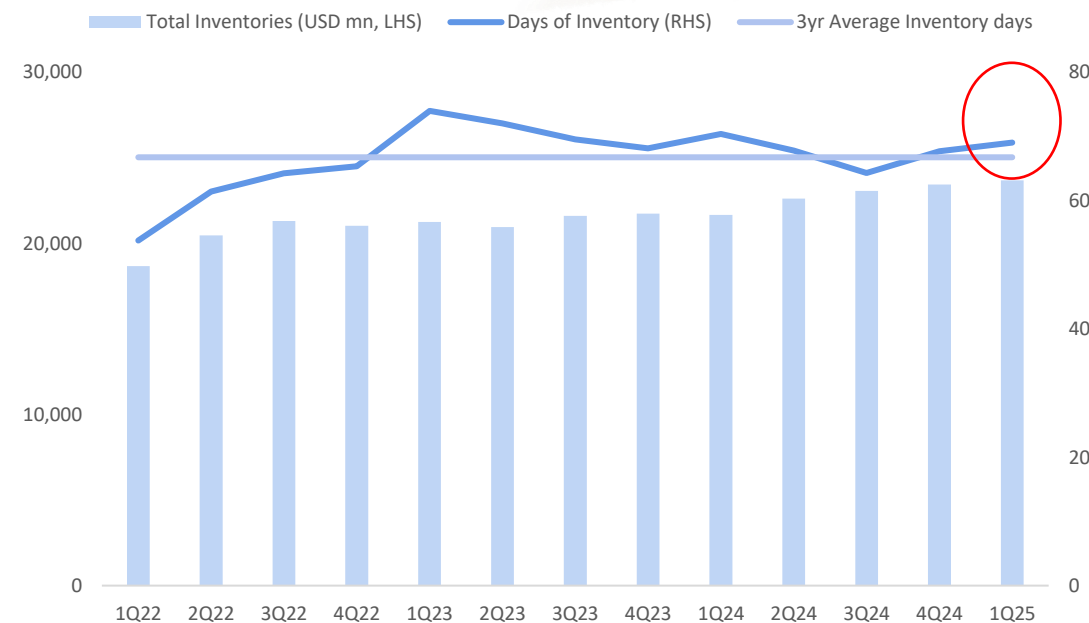
半导体库存仍可控，但开始上升

TOP-15 Fabless DOI



数据包含: NVDA, AMD, AVGO, QCOM, MTK, MRVL, QRVO, Novatek, Realtek, SWKS, HIMX, WILL, MPWR, IFX, NXP

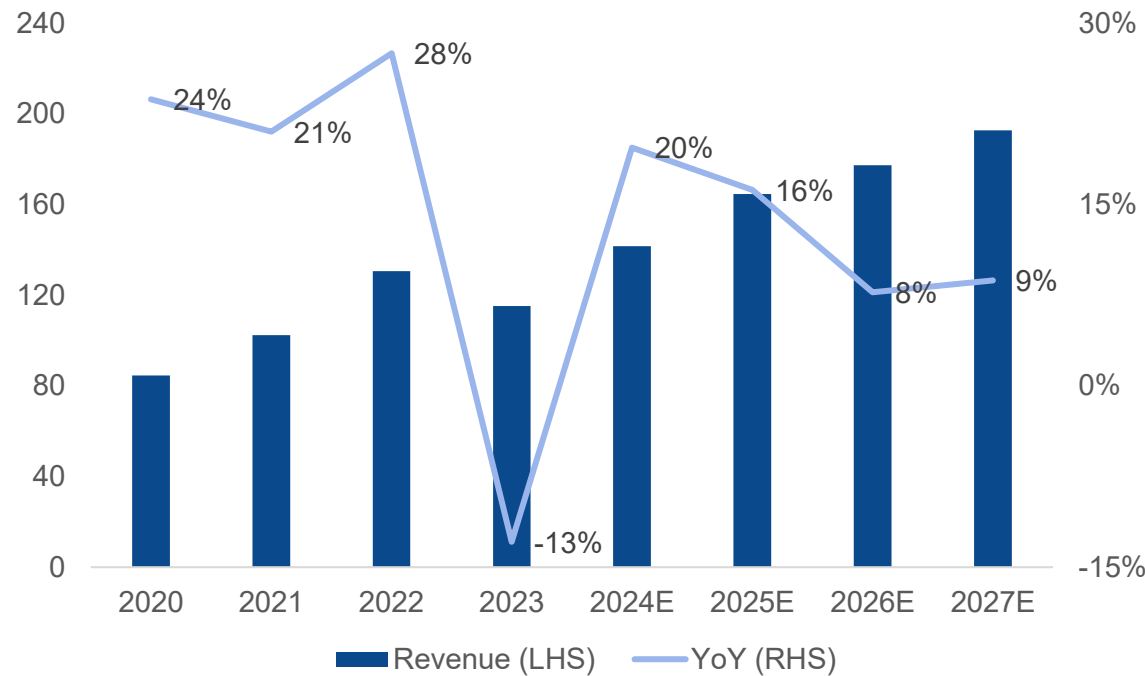
Distributor DOI



数据包含: ARW, WPG, AVT, WT Micro, Macnica, CECport, Supreme, Edom

晶圆代工市场受益AI和Intel委外

全球晶圆代工 (1.0) 市场 (USD, bn)

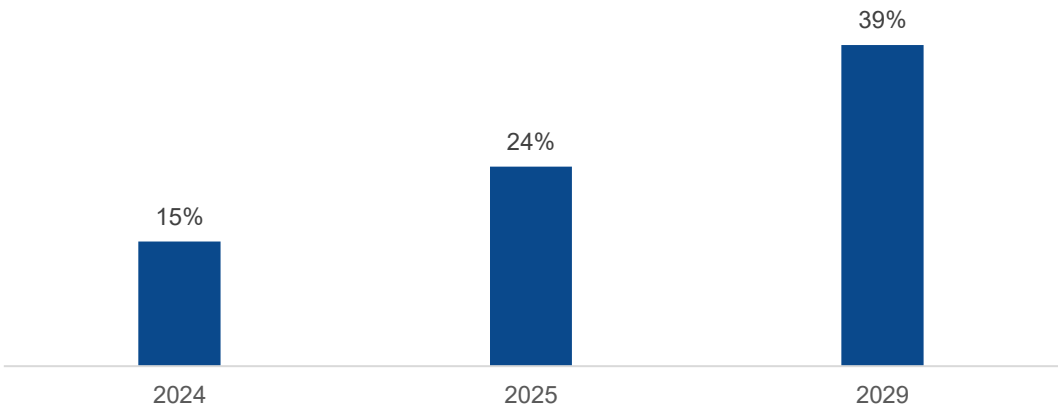


资料来源: Company data, GFHK

Intel PC CPU 委外

		CPU SKU at Intel	CPU	CPU SKU to TSM	CPU
2024	Meteor Lake	P, M	Intel 4		
2H24	Arrow Lake	P	Intel 20A	SK (DT), S (DT), P	TSM N3B
2H24	Lunar Lake			M	TSM N3B
2H25	Arrow Lake			S refresh	TSM N3B
2H25	Panther Lake		Intel 18A		
2H26	Nova Lake	S, P, M(?)	Intel 18A	S, P	TSM N2P
2H27	Razor Lake	S, P, M(?)	Intel 14A	S, P	TSM N2P

台积电预测AI收入贡献



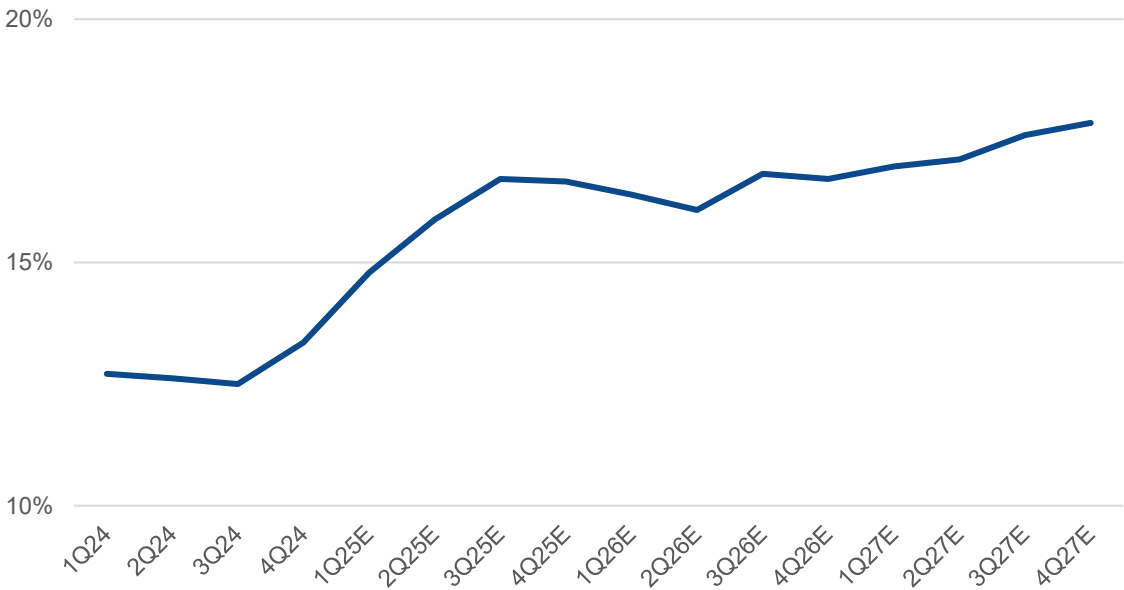
台积电 vs. 地缘政治，先进制程美国占比长期30%

- 台积电累积已公告USD165bn投资额于美国，预计先进制程产能30%于美国，毛利影响扩大

台积电美国厂量产节奏

Fab	Process Technology	Expected Production Start
Fab 1	N4	4Q24
Fab 2	N3	2027
Fab 3	2nm or more advanced node	End of the decade
Fab 4, 5, 6		
Packaging 1, 2		

台积电海外产能占比温和上升



资料来源: Company data, GFHK

半导体设备支出走向持平或衰退

- 2026先进制程支出下滑，AMSL EUV订单量下行

全球WFE模型 (USD bn)							
Billion USD	2020	2021	2022	2023	2024	2025E	2026E
IDM	11	12	16	17	17	15	12
Memory	27	39	39	24	33	37	39
Foundry	27	42	49	42	45	49	49
World Total	65	93	104	83	96	101	101
Total YoY	16%	44%	12%	-20%	15%	5%	0%

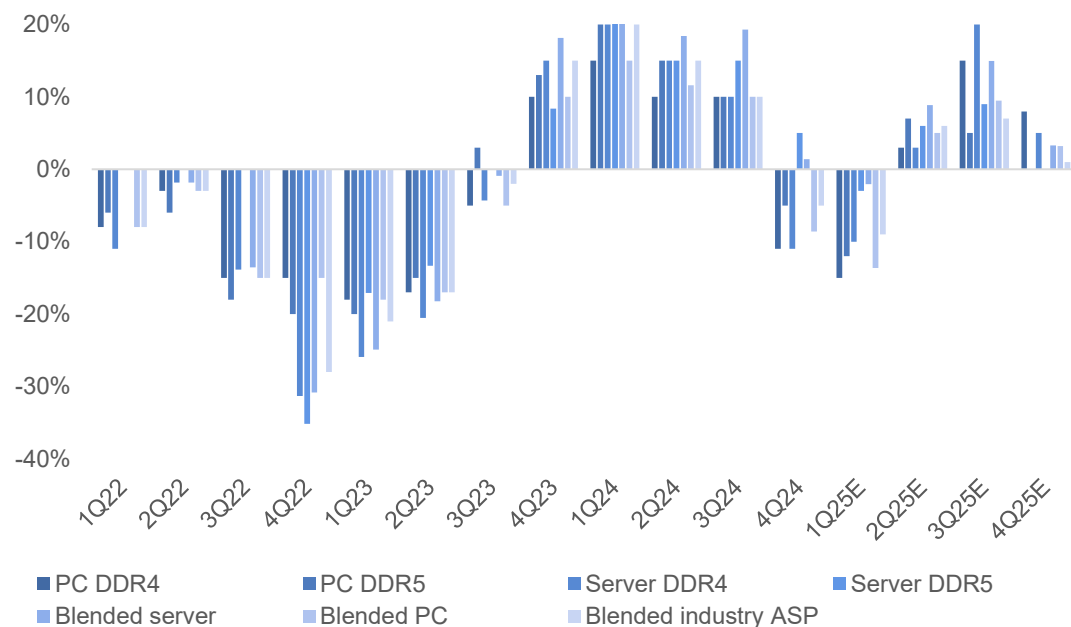
ASML Low NA EUV需求预测			
	FY25E	FY26E	FY27E
TSMC	16	14	20
Samsung	10	12	14
Intel	10	5	8
Micron	4	4	5
Hynix	6	6	4
Others	2	2	1
Total	48	43	52

资料来源: Semi, GFHK

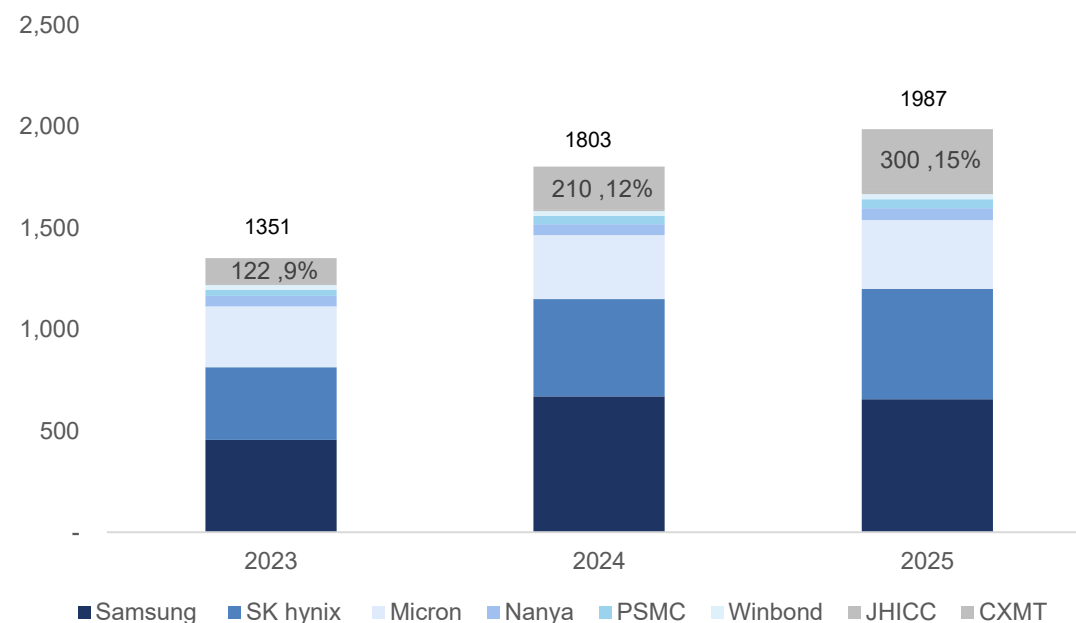
存储短期价格略超预期，然上行周期进入尾声

- 3季度DDR5涨幅或略超预期，因供应链混乱和服务器需求强劲
- 预期4Q25 DDR5价格将开始走平，伴随国内全面转向DDR5，2026年价格压力增大

DRAM合约价环比



CX 产能 (KPM, %)



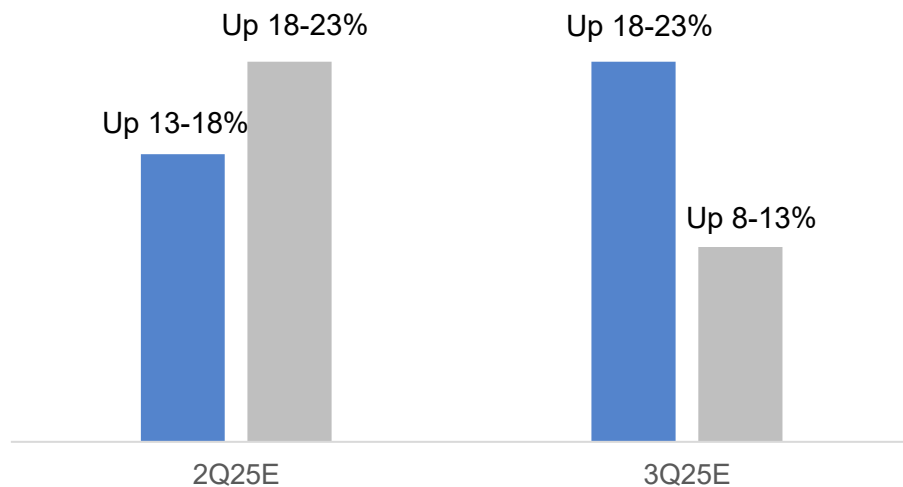
资料来源: TrendForce, GFHK

利基型存储受益大厂退出和手机3D DRAM

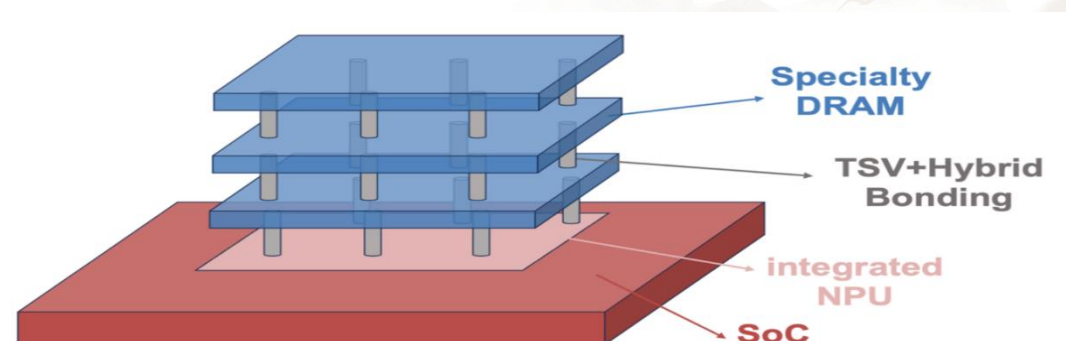
- 高通将于2H26年采用3D-DRAM，旗舰SoC新设计提升至200个Tokens。3D-DRAM看好兆易创新（603986 CH），和南亚科（2408 TT）

DDR4价格涨幅预测

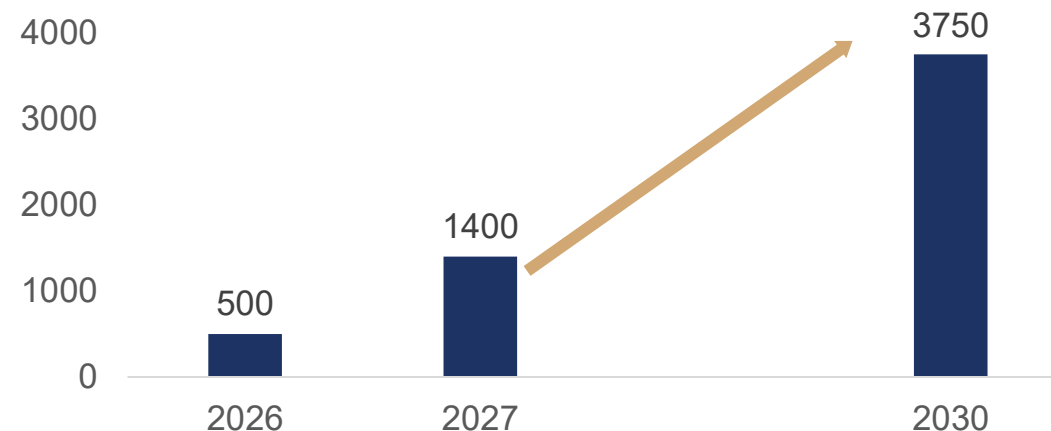
■ PC DDR4 ■ Server DDR4



手机端3D DRAM



手机端3D DRAM市场规模



CSP 资本支出增速或优于预期

- 微软为首家CSP给出26年指引 (FY26)，我們预期\$90bn以上，对应FY25增速10%+，合计北美4大CSP +15% YoY

北美CSP Capex

(USD bn)	2022	2023	2024	2025E	2026E
MSFT	28	41	76	92	103
AMZN	64	53	83	106	
Google	31	32	53	77	
Meta	32	28	39	68	
Top 4	155	154	250	344	
YoY	17%	-1%	62%	37%	15%+

中国CSP Capex

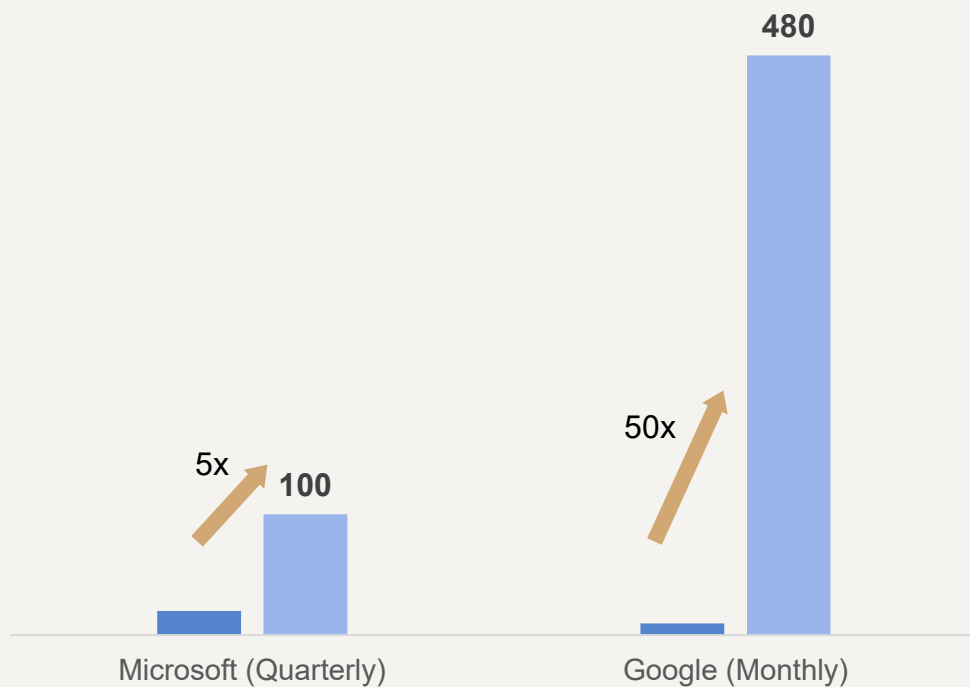
(RMB bn)	2022	2023	2024	2025E
腾讯	18	24	77	70
百度	8	11	8	12
阿里	48	24	76	120
字节			80	140
Top 4	75	59	241	358
YoY	-16%	-21%	308%	48%

Tokens、AI ARR 增长强劲

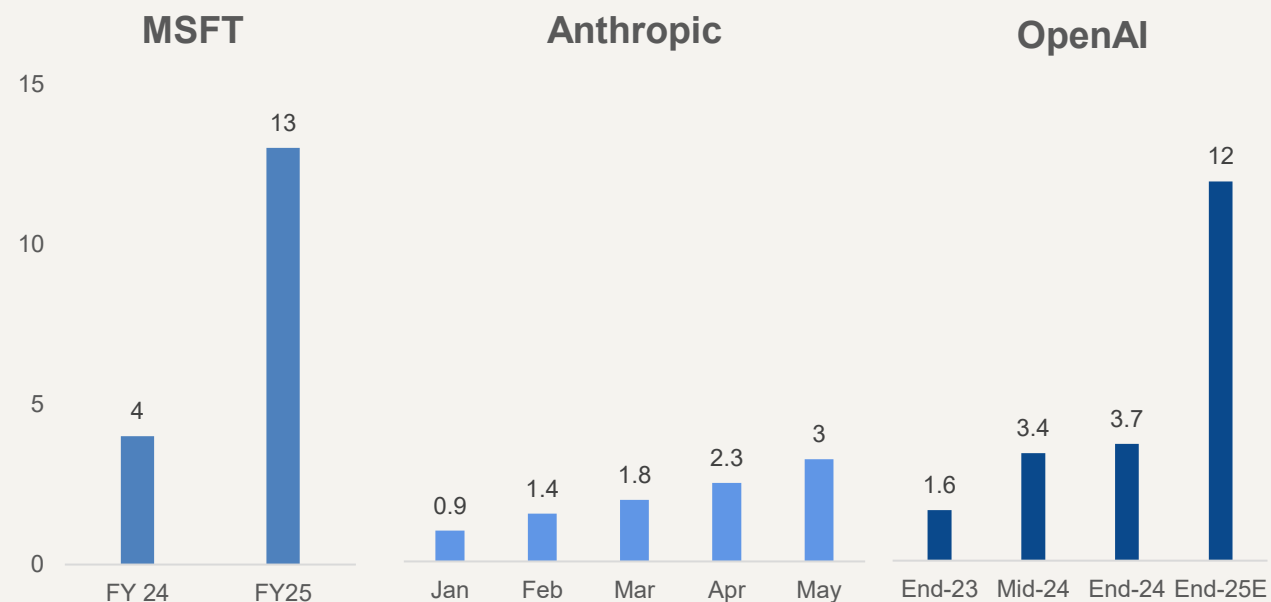
- 微软、Google近期公告tokens处理数量大增，显示推理的应用大幅增长

北美CSP公告token数

■ A year ago ■ Now



AI ARR (USD bn)



资料来源: Company, Benzinga, New York Times

AI加速器市场规模

- 预期GPGPU/NVDA、ASIC 同时增长，GPU仍有较大的性能和使用场景的优势
- 我们于5月9日大幅上修ASIC明年预期 > 博通、CLS、光模块、CCL

资料来源: Company data, GFHK

AI加速器市场测算

	2024	2025E	2026E	2027E
GPGPU (Units m)				
NVIDIA	3.9	5.9	7.7	7.2
AMD	0.5	0.6	0.8	0.9
Others	0.7	1.3	1.9	3.4
Total	5.1	7.9	10.4	11.5
YoY	160%	55%	32%	11%
ASIC (Units m)				
Google	2.8	3.0	3.3	4.3
AWS	0.9	1.7	2.3	3.1
Meta	0.2	0.5	0.8	1.3
MSFT	0.1	0.0	0.4	1.0
Other ASICs	0.3	0.8	1.0	2.0
Total	4.2	6.0	7.8	11.6
YoY	125%	43%	30%	50%
Market (USD bn)				
NVIDIA	88	156	199	231
Total GPGPU	100	179	234	292
Total ASIC	23	32	52	89
Total Market	123	211	286	381
YoY	175%	72%	35%	33%

NVDA 收入模型

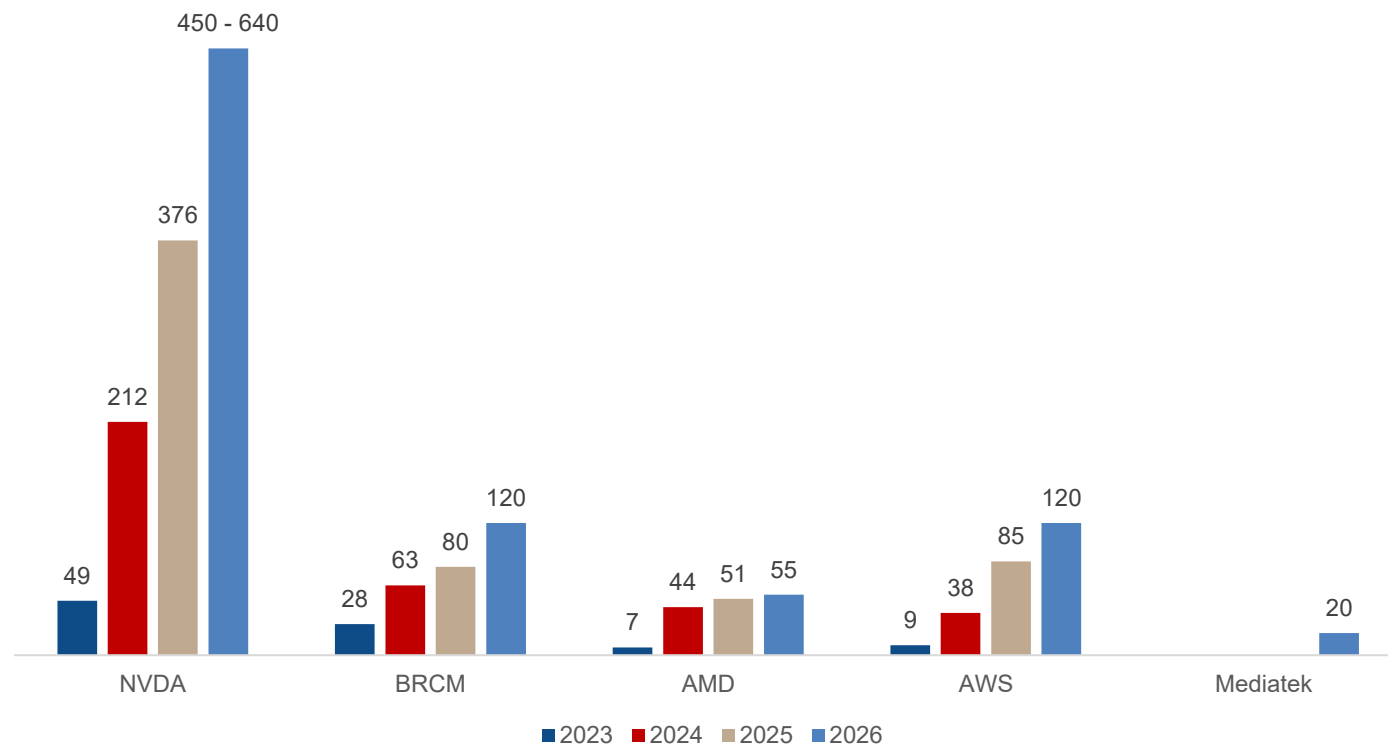
- 公司给出July-Q指引\$45bn, 略低于广发\$46bn, 然高于华尔街\$44.5bn
- 预期强劲的在手订单支持下 一季度指引

Nvidia 收入模型											
FY	2024	1Q25	2Q25	3Q25	4Q25	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E
CY		Apr-24	Jul-24	Oct-24	Jan-25		Apr-25	Jul-25	Oct-25	Jan-26	
AI GPU (k units)											
A100	507	30				30					0
H100/H200/H20	899	610	816	1,196	974	3,596	500	150			650
B200	-	-	-	-	300	300	810	1,018	1,200	1,350	4,378
6000D	-	-	-	-	-				300	600	900
R100	-	-	-	-	-						
Total	1,405	640	816	1,196	1,274	3,926	1,310	1,168	1,500	1,950	5,928
Revenue (USD mn)											
Datacenter	47,503	22,563	26,272	30,771	35,580	115,186	39,112	40,640	47,276	55,366	182,393
Total	60,922	26,044	30,040	35,082	39,331	130,497	44,062	46,513	53,275	60,433	204,284
NV Guidance		24,000	26,000	32,500	37,500		43,000	45,000			
GF Preview					38,000		42,500	44,000			

CoWoS 产能跟踪

- 英伟达杂音多，CoWoS较年初下修，然近月的调整微调向下
- AMD MI350需求强劲，然其CoWoS产能充裕，短期无须增加
- 博通于近一个月有所上调
- AWS于近一个月有所上调

CoWoS产能(000, 年)

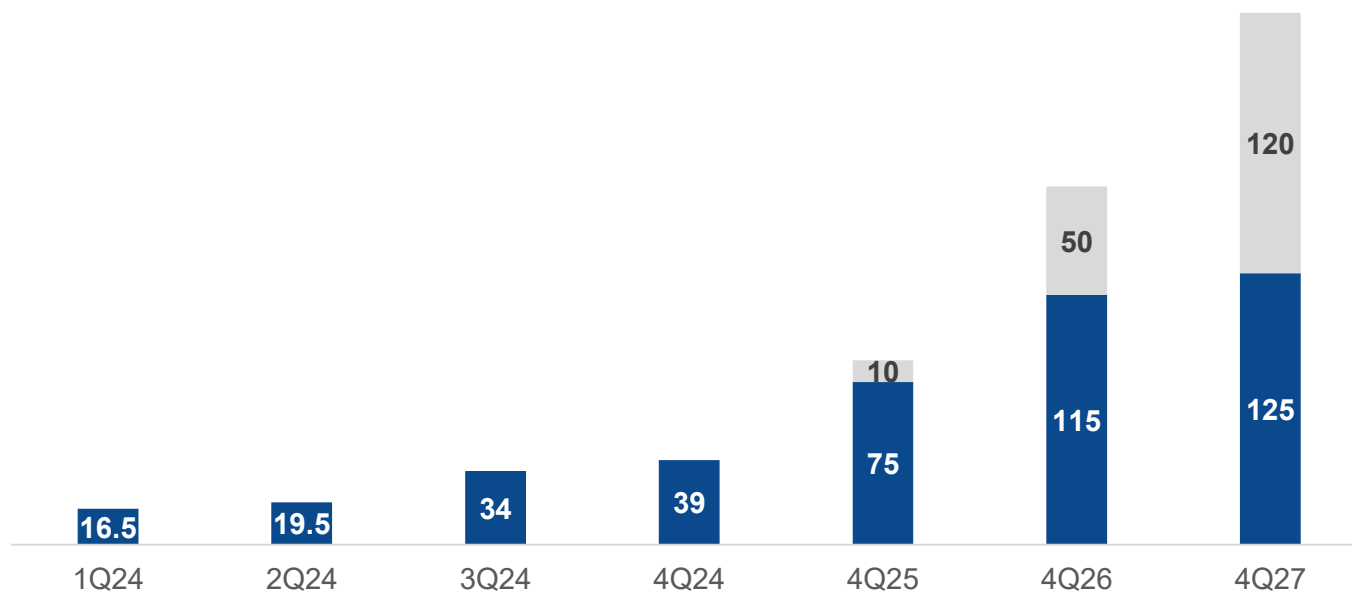


CoWoS、WMCM 产能积极扩张

- iPhone 18 Fold、18 Pro Max 和 18 Pro 将搭载 A20 芯片，采用台积电的晶圆级多芯片模组 (WMCM) 封装和 N2
- CoWoS 需求因 ASIC 增长而激增，我们仍预测台积电的 CoWoS-L+S 产能将从 2025 年底的 75KPM 增至 2026 年底的 110KPM。
- CoWoS 设备受益者— ONTO (Buy)、弘塑 (3131 TT)、Shibaura (6950 JP)

TSM CoWoS 产能 + TSM WMCM (000)

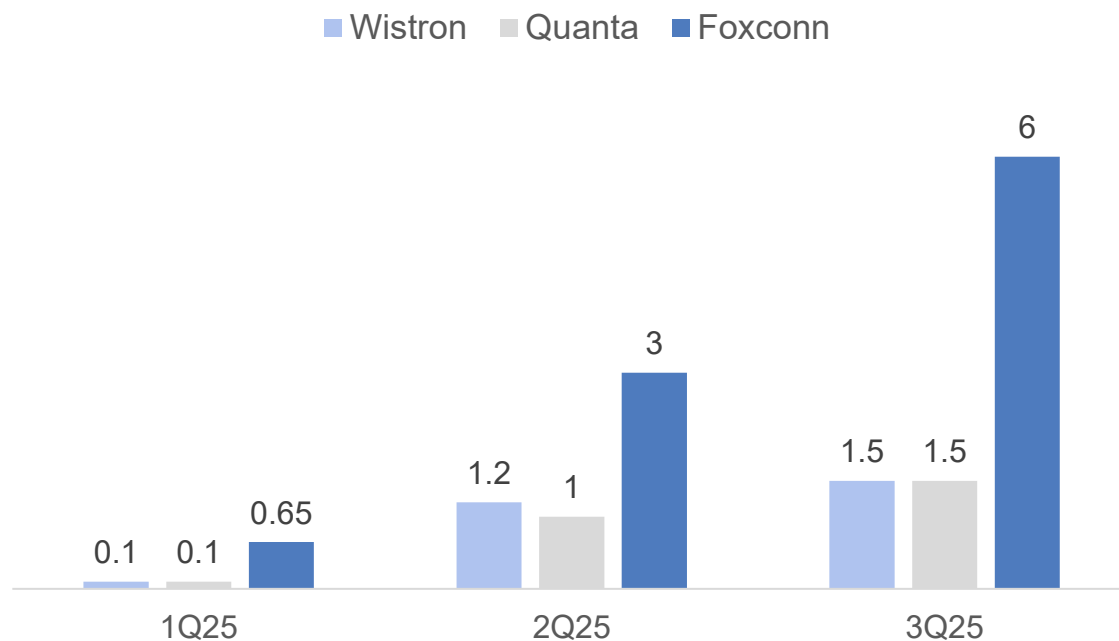
■ Total CoWoS ■ WMCM



GB200直通率自4月起改善

- 受诸多因素影响，GB200上量时间推迟，客户1-2季度转HGX或GB300
- 4月已改善Firmware (1.0版本释出) 和美国L10自动化，且L11 Cycle time已有所进步

GB200 出貨預估 (NVL 72, 000)



GB200+GB300 預估 (NVL 72, 000)

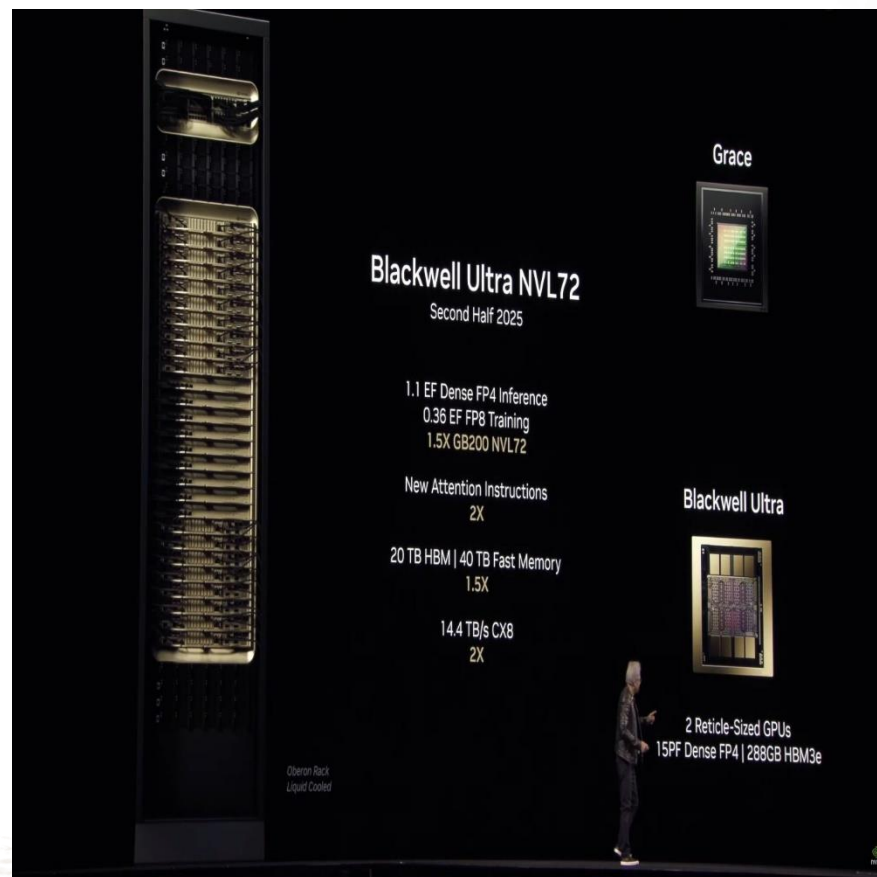
	2024-2025	GB200 + GB300
MSFT	6	3.5 + 2.5
Meta	3	
Tesla	4	
AWS	1	
GOOG	4	
Oracle	6	
Coreweave	1	
Others	1	
Total	25	18 + 7

GB300 (Ultra) NVL72 设计变更

GB300 Compute Board 或退回Bianca

	GB200 Module x2 (Bianca)	原GB300 Module x1 (Cordelia)
Compute node	(1CPU+2GPU) x2	(2CPU+4GPU) x1
CPU	Soldered down	Soldered down
GPU	Soldered down	SXM Module
CPU memory	LPDDR5X, Soldered down	SOCAMM module
NV Link	18 NVL Ports/ per GPU	18 NVL Ports/ per GPU
Scale Out	CX7 I/O board	CX8 I/O module
Management	BMC + HMC	BMC + HMC (Optional)
Input voltage	12V via bus bar	54V via bus bar
PDB	Single PDB Board	Dual PDB Board
Cold plate	2	6 + 1-2 (fanless)
QD	6	14
PCB	HDI	UBB (M8+M4)

GB300 NVL72 于GTC发表



- **Compute board 退回 Bianca**、CPO (选配)、BBU (选配)、电容 (超级电容选配)
- 3月测试, **量产提早至7-8月**

NVDA 路标图，6000D进度有待观察

GTC – GPU 路标图

	H100	H200	B200	B300 Ultra	B300A	Rubin	Rubin Ultra
Launch	4Q22	2Q24	4Q24	3Q25	3Q25	2H26	2H27
Architecture	Monolithic	Monolithic	Dual-die	Dual-die	Single-die	Dual-die + 2x I/O	4-die + 2x I/O
Front-end node	N4P	N4P	N4P	N4P	N4P	N3P	N3P
Back-end process	CoWoS-S	CoWoS-S	CoWoS-L	CoWoS-L	CoWoS-L	CoWoS-L	CoWoS-L
Transistor (bn)	80	80	208				
Interconnect	NV Link 900GB/s	NV Link 900GB/s	NV Link 1,800 GB/s	NV Link 1,800 GB/s	NV Link 1,800 GB/s	NV Link 3,600 GB/s	NV Link 3,600 GB/s
HBM (GB)	80GB HBM3	141GB HBM3E	192GB HBM3E	288GB HBM3E 12 hi x 8	144GB HBM3E 12 hi x 4	288GB HBM4 12 hi x 8	1024GB HBM4E 12 hi x 16
Memory Bandwidth (TB/s)	3.35	4.8	8	8	4	13	32
TDP	700W	700W	1000W	1000W	600W	1,800W	3,600W
Networking	400Gb CX7	400Gb CX7	400Gb CX7	800Gb CX8	800Gb CX9	1,600Gb CX9	1,600Gb CX9

资料来源: Company data, GFHK

GTC 路标图以外

	H20	6000D (aka B40)
Launch	2Q24	July-2025
Architecture	Monolithic	GB202
Front-end node	N4P	N4P
Back-end process	CoWoS-S	CoWoS-L
Transistor (bn)		
Interconnect	NV Link 1,800GB/s	PCIe
HBM (GB)	96/141GB HBM3	80GB GDDR7
Memory Bandwidth (TB/s)	4.0	1.7
TDP	400W	600W
Networking		

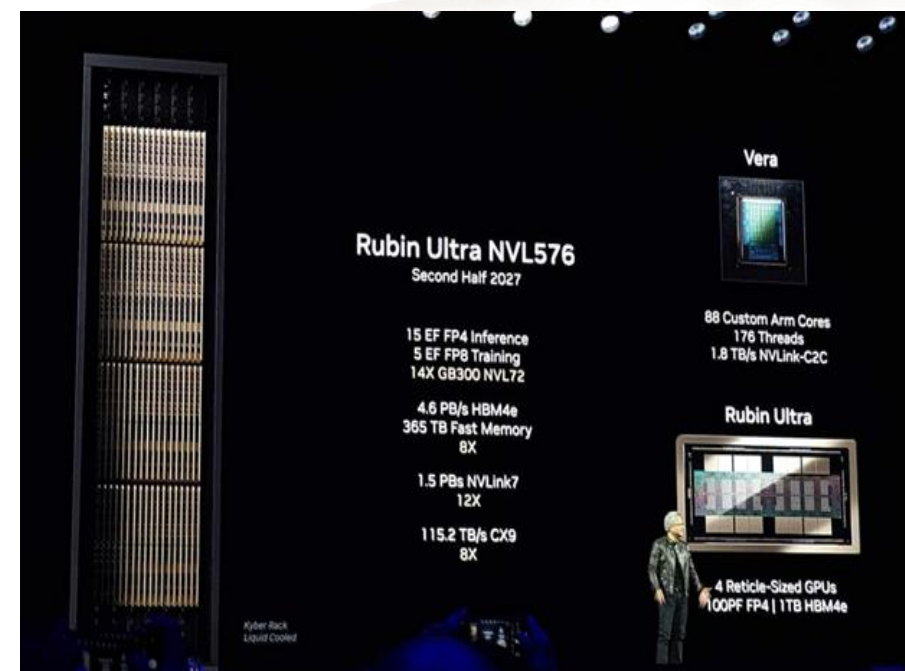
Rack Level NVL576 于 2027年

GTC – Rack Level 路标图

	B200 NVL8	GB200 NVL72	GB300 NVL72	B300 NVL16	VERA RUBIN NVL 144	VERA RUBIN Ultra NVL576
Schedule	4Q24	4Q24	3Q25	3Q25	2H26	2H27
Form Factor (CPU:GPU package)	2 : 8	2 : 4	2 : 4	2 : 16 (B300A die)	2 : 4	1 : 1
Cooling system	Air or Liquid	Liquid	Liquid	Air or Liquid	Liquid	Liquid
GPU package	2 die, 8hi x 8	2 die, 8hi x 8	2 die, 12hi x 8	1 die, 12hi x 4	2 die, 12hi x 8	4 die, 12hi x 16
GPU chip/package volume	8	36 / 72	72	16	72	144
HBM (GB) per GPU	192GB HBM3e	192GB HBM3e	288GB HBM3e	144GB HBM3e	288GB HBM4	1024GB HBM4E
Memory Bandwidth (TB/s)	8	8	8	4	13	32
GPU TDP	700/1200W	1200W	1400W	600W	1800W	3600W
NVLink Bandwidth (GB/s)	1800	1800	1800	1800	3600 (4 switch AISC/tray)	3600
NVLink GPU memory (TB)	1.5	14	21	2.3	21	147
Networking Adapter	400Gb CX7	400Gb CX7	800Gb CX8	800Gb CX8	1600Gb CX9	1600Gb CX9

资料来源: Company data, GTHK

Rubin Ultra NVL576



- 2027年NVL576架构受益CCL (PTFE、M8)、PCB、电源 (~600W)、散热

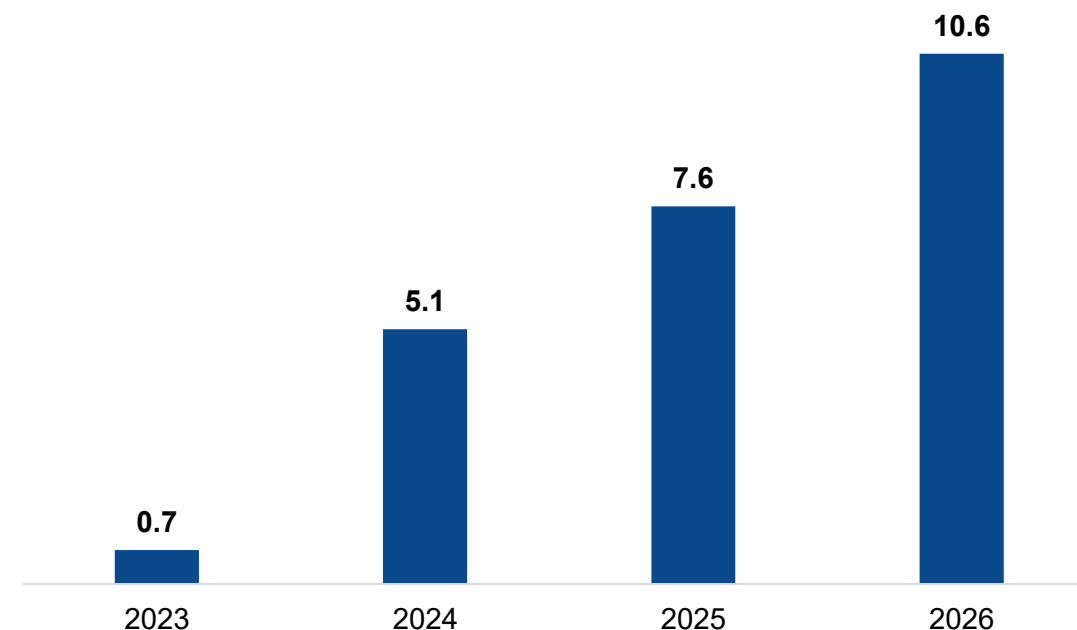
AMD MI355 需求优于预期

- 下半年MI350/MI355已经取得主要CSP和企业订单，主要由Meta、戴尔与甲骨文采用
- 预期明年MI450竞争力大增，因互连能力进步和Rack系统架构IF128

AMD AI GPU Roadmap

	MI300A	MI300X	MI325X	MI355X	MI450X
Launch	4Q23	4Q23	4Q24	3Q25	2H26
Front-end node	N5	N5	N5	N3P	N3P
Back-end process	SolC+	SolC+	SolC+	SolC+	SolC+
	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-L
Transistor (bn)	146	153	153		
Interconnect (per chip, unidirectional)	128GB/s	128GB/s	128GB/s	128GB/s	~1,600GB/s
HBM (GB)	HBM3	HBM3	HBM3e	HBM3e	HBM4
	128GB	192GB	256GB	288GB	288GB
Memory Bandwidth (TB/s)	5.3	5.3	6	8	
TDP	750W	750W	1000W	1000W	1,600-2,000

AMD AI 收入 (USD bn)



ASIC项目跟踪

- 博通为最领导厂商，然联发科崛起，关注TPU v7e进度
- AWS 延长R2生命周期，T3两案 (Alchip、MRVL) 并行，MRVL项目预计于3Q25 tape-out
- Meta 明年ASIC改为HBM将增加其竞争力

2026-2027主要ASIC

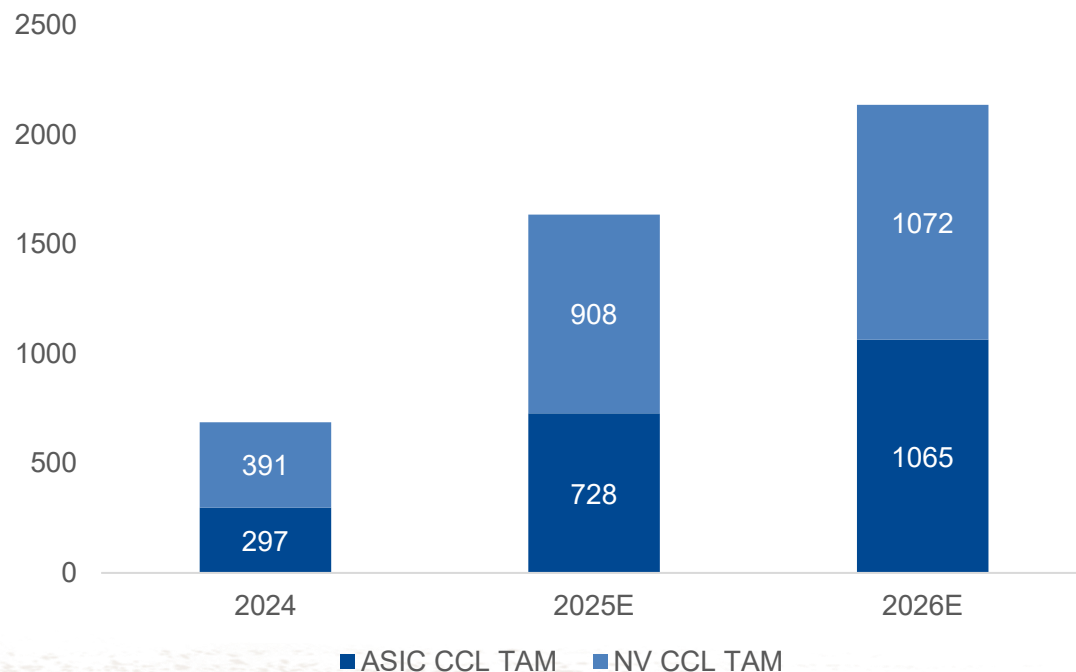
	TPU v7p	TPU v7e	TPU v8p	TPU v8e	MTIA v1	MTIA v1.5	AWS R2	AWS T3	AWS T3 Lite	MSFT M200	MSFT M300	MSFT M400	OpenAI	OpenAI	Apple
时间	3Q26	End-26	3Q27	3Q27	1H26	2Q26	4Q25	2Q26	3Q26	4Q25	2H26	2H27	2H26	2027	2027
设计厂商	BRCM	GOOG, MTK	BRCM	GOOG, MTK	BRCM	BRCM	MRVL	Annapura, Alchip	MRVL	GUC	MRVL	-	BRCM	BRCM	BRCM
前道制程	N3	N3	N2	N2	N3	N3	N5	N3	N3	N3	N2	N2	N3	A16	TBD
Memory	HBM3e 12hi	HBM3e 12hi	HBM4	HBM4	HBM3e	HBM3e 12hi	HBM3e 12hi	HBM3e 12hi		HBM3e	HBM4	HBM4	HBM3e	SRAM	TBD
后道制程	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-R	CoWoS-R	CoWoS-R	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-R	CoWoS-R	TBD

资料来源: Company data, GFHK

ASIC带动高阶CCL需求

- ASIC PCB及CCL规格高于GPU，我们预期AI server CCL TAM 2026年达到2.1bn USD
- 下一个关注点为Nvidia Rubin系列是否会采用M9材料

CCL TAM预测 (mn USD)



各等级CCL价格预估 (USD)

M6	37
M7	73
M8 (Low Dk 1)	150
M8 (Low Dk 2)	250
M9	330

ASIC PCB/CCL供应链

各家ASIC PCB/CCL供应情况

	NVIDIA			Google					AWS			Meta	
芯片	B200	GB200	GB300	TPUv5p	TPUv5e	TPUv6p	TPUv6e	TPUv7p	T2	T2.5	T3	MTIA2	MTIA3
PCB spec	24L PTH (M8)	22L HDI (M8+M4) 26L PTH (M8)		24L PTH (M7)	22L PTH (M7)	36L PTH (M8+M6)	34L PTH (M8+M6)	TBD	26L PTH+HDI (M8)		32L PTH+HDI (M8)	32L PTH (M8+M2)	32L PTH (M8+M2)
单芯片PCB价值量 (RMB)	2,000	2,500	2,500	2,500	2,000	3,500	2,800	TBD	5,000	5,000	4,500	6,500	6,500
PCB供应商	胜宏、欣兴、沪电			ISU、沪电 TTM、深南		ISU、沪电、TTM 胜宏（新）			生益、GCE TTM、高技		生益、GCE TTM、高技	沪电、TTM 深南、ISU	沪电、TTM 深南 广和/GCE
CCL供应商	斗山、台光			松下		松下		TBD	台光		台光	台光	

光模块 TAM, 800G上修

800G 光模块预测

(m units)	2022	2023	2024	2025E	2026E
800G	0.1	2.0	10.0	17.0	37.6
Google	0.1	1.0	3.5	3.5	8.0
AWS				2.5	6.0
Meta		0.2		4.0	8.0
NV		0.8	6.0	3.0	3.0
MSFT				2.0	4.0
Alibaba				0.4	0.5
Tencent				0.4	0.5
Huawei				0.2	0.6
Others			0.5	0.2	0.2

1.6T 光模块预测

(m units)	2023	2024	2025E	2026E
1600G		0.2	1.5	4.0
Google		0.1	0.4	0.7
AWS				0.1
Meta				0.1
NV		0.1	1.1	2.5
MSFT				0.1
Alibaba				
Tencent				
Huawei				
Others				0.5

Nvidia 首款CPO交换机需求不佳

- 3Q25推出CPO IB交换机 (备货从10k下修至2k)，放量看2H26的以太，明年看10k-20k

Nvidia CPO 交换机

产品	Taipan QM3-CPO	SPC5-CPO	SPC6-CPO
起量时间	3Q25	2026	2026
交换芯片	Quantum 3	Spectrum 5	Spectrum 6
激光器	Lumentum	Lumentum	Lumentum
衬底	UMTC	UMTC	UMTC
FAU供应商	TFC	TFC/Senko/Corning	TFC/Senko/Corning
模组组装厂	Fabrinet	Fabrinet, SPIL	SPIL
封装厂	Amkor	TSMC	TSMC/SPIL

CPO 交换机供应链预测

- 英伟达现有的CPO方案是用Mellanox的供应链，但后续关注台积电供应链

NV Mellanox 供应商

器件	供应商
FAU	天孚, Senko, 康宁
Laser	Lite
ELS module	昂纳, 天孚, 旭创、新易盛
模组	Fabrinet
衬底	Unimicron
Fiber Shuffle	康宁, 太辰光
ABF	欣兴, Ibiden
多层板	WUS, 胜宏, 方正

NV TSMC 供应商

器件	供应商
FAU	上詮、天孚, Senko, 康宁
Laser	Lite
ELS module	昂纳、天孚、旭创、新易盛
模组	SPIL
衬底	Unimicron
Fiber Shuffle	康宁, 太辰光
ABF	欣兴, Ibiden
多层板	WUS, 胜宏, 方正

博通 CPO供应商

器件	供应商
FAU等光器件	Senko
Laser	AVGO
ELS module	众达
模组	讯芯
衬底	Nanya PCB
EIC/PIC design	AVGO
EIC/PIC foundry	Silterra/Tower
EIC/PIC OSAT	ASE

智能机出货预期

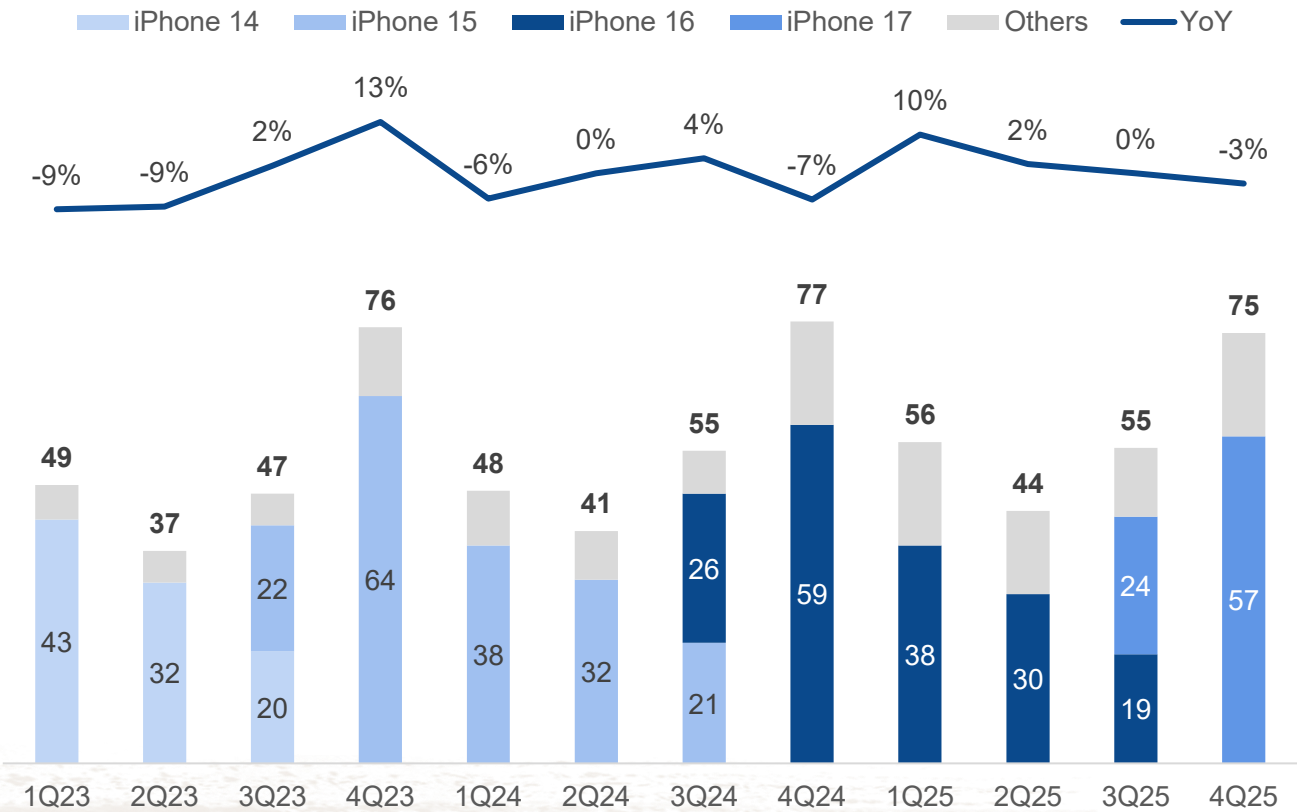
全球智能机和主要品牌出货预估 (百万台)

Shipments (mn)	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2024	2024 YoY	2025E	2025 YoY
Global smartphone	268	268	304	324	300	292	316	332	305	1240		1255	
YoY	-14.5%	-6.3%	0.6%	6.8%	11.9%	8.9%	4.0%	2.3%	1.5%	6.5%		1.2%	
By OEMs													
Samsung	60.5	53.5	59.5	53.0	60.1	53.8	57.8	51.7	60.6	223	-1%	218	-2%
Apple	55.4	44.5	54.1	80.2	52.6	45.7	56.0	76.9	57.9	231	-1%	232	0%
Huawei	6.5	9.6	8.1	10.6	12.5	13.7	11.2	11.0	11.0	48	39%	50	3%
New Honor	12.8	13.8	17.7	17.1	16.8	15.3	15.0	17.0	15.0	64	4%	65	2%
OPPO/Realme	35.6	35.5	37.8	35.4	35.3	38.1	42.8	39.1	35.5	155	8%	165	6%
vivo	20.6	21.3	22.0	24.0	21.3	25.9	27.0	26.0	22.7	100	14%	105	5%
Xiaomi	30.5	33.2	41.5	40.7	40.8	42.3	42.8	42.7	41.8	169	15%	178	6%
Share %													
Samsung	23%	20%	20%	16%	20%	18%	18%	16%	20%	18%		17%	
Apple	21%	17%	18%	25%	18%	16%	18%	23%	19%	19%		18%	
Huawei	2%	4%	3%	3%	4%	5%	4%	3%	4%	4%		4%	
New Honor	5%	5%	6%	5%	6%	5%	5%	5%	5%	5%		5%	
OPPO/Realme	13%	13%	12%	11%	12%	13%	14%	12%	12%	13%		13%	
vivo	8%	8%	7%	7%	7%	9%	9%	8%	7%	8%		8%	
Xiaomi	11%	12%	14%	13%	14%	14%	14%	13%	14%	14%		14%	

资料来源: IDC, GFHK

iPhone 需求平淡

iPhone 组装量预估(百万台)



资料来源: Company data, IDC

iPhone 16e性价比不佳

	iPhone SE3	iPhone 16e	iPhone 16
Launch	Mar 2022	Feb 2025	Sep 2024
Price	\$429 - \$579	\$599 - \$899	\$799 - \$1099
Processor	A15 Bionic	A18	A18
Apple Intelligence	No	Yes	Yes
Rear Camera	Single-Camera 12MP	Single-Camera 48MP Fusion 2x Telephoto	Dual-Camera 48MP Fusion 12MP Ultra-wide
Front Camera	7MP	12MP	12MP
Display	4.7-inch LCD	6.1-inch OLED	6.1-inch OLED
Modem	Qualcomm X57 5G	Apple C1 cellular modem	Qualcomm X71 5G
RAM	4GB	8GB	8GB
Connector	Lightning	USB-C	USB-C
Face ID	No	Yes	Yes
Action Button	No	Yes	Yes
Battery Life	Up to 15 hours	Up to 26 hours	Up to 22 hours
Charging	Qi Wireless	Qi Wireless	Magsafe & Qi Wireless

iPhone 17展望平平，待26年摺疊产品、AI

iPhone 17规格预测

	iPhone 17	iPhone 17 Air	iPhone 17 Pro	iPhone 17 Pro Max
Launch	Sep-25	Sep-25	Sep-25	Sep-25
Display	6.1"	6.6"	6.3"	6.9"
Processor	A18 N3E	A19 N3P	A19 Pro, N3P	A19 Pro, N3P
DRAM	LPDD5 8GB	LPDD5 12GB	LPDD5X 12GB	LPDD5X 12GB
Front camera	24MP, 6P	24MP, 6P	24MP, 6P	24MP, 6P
Rear camera	48MP 7P, 12MP 5P	48MP 7P	48MP 7P, Periscope 48MP 1G+3P, 48MP 6P	48MP 7P, Periscope 48MP 1G+3P, 48MP 6P
Face ID	Structured light	Structured light	Structured light	Structured light
Casing	Aluminum	Titanium	Aluminum	Aluminum
BB	Qualcomm	Qualcomm or Apple	Qualcomm	Qualcomm
Charging	35W	35W	35W	35W
NPI	Foxconn	Foxconn	Foxconn	ICT

折叠产品

	Foldable IT Product	Foldable iPhone
Display	18.8"	7.74" (active area) (Sub: 5.49 or 6.55)
Panel	LG Display	Samsung, LG Display
Hinge	SZS	Amphenol, SZS
Touch Solution	On-cell Touch	On-cell Touch
Product life cycle	2 - 3mn	16-18mn
Mass Production	4Q26	4Q26
NPI	Jan 2025, Foxconn	Jan 2025, Foxconn

资料来源: Company data, MacRumors

Apple Intelligence 不确定性

- 苹果公司证实，原定于2025年推出的Apple Intelligence Siri 延后至2026年
- 苹果硬件建置前期非常缓慢 (i.e., 服务器、ASIC)，然**近期已开始采购Nvidia方案**

iOS 版本更新

	iOS 18	iOS 18.1	iOS 18.2	iOS 18.3	iOS 18.4
Release Time	2024/9/16	2024/10/28	2024/12/11	2025/1/27	April 2025
Features	No Apple Intelligence	- Writing tools - Smart replies - Summaries - Focus mode	- Genmoji - ChatGPT - Visual intelligence - Image AI	- Notification upgrade - Default Apple intelligence	- New Emoji - Default Map and Translation in EU - Robot Vacuum Support
Service Region	N/A	US English Only	More English countries	No change	Non-English countries

资料来源: Company data, SemiAnalysis, GFHK

Apple 芯片和其他产品推进

- 基带C1或导入iPhone Air、 C2 iPhone 18 P/PM
- 自研WiFi 7于25年下半年
- 配件较无亮点， AR眼镜仍不明确

苹果自研芯片					
	1H24	2H24	1H25	2H25	1H26
A series	A18/A18 Pro (N3E)		A19/A19 Pro (N3P)		A20/A20 Pro (N2)
M series	M4 (N3E)	M4 Pro/Max (N3E)	M4 Ultra (N3E)		M5 Pro/Max (N3P)
			M5 (N3P)		M6(N2)
Others		Modem (N4)	Wi-Fi (N6)		R2 (N2)

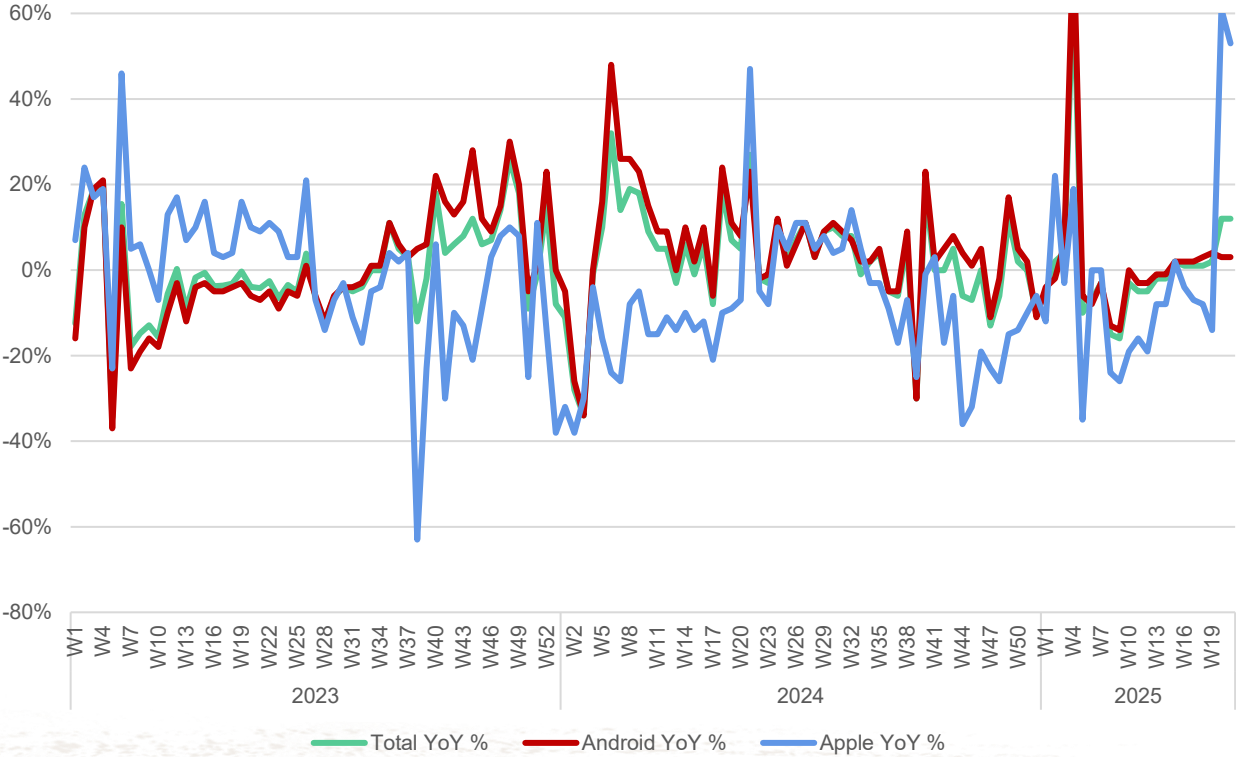
苹果其他产品

	2025	2026	2027~
AirPods	AirPods Pro 3		
Apple Watch	Apple Watch S11 series Apple Watch Ultra 3	Apple Watch S12	Apple Watch featuring Blood monitoring
Others	HomePad 6"	Vision Pro refreshed (TBD)	Vision Air (TBD) Smart Glass (TBD)

资料来源: Company data, SemiAnalysis, GFHK

国内手机於五一销售略有好转，后续看618

国内手机周度销量YoY



Android手机接入AI

	Third Party models	Own AI model
Samsung	Google Gemini Baidu (China)	Gauss
Huawei	N/A	Pangu
Honor	Baidu 千帆, Bytedance 豆包, DeepSeek	MagicLLM
OPPO	Google Gemini, Baidu, DeepSeek	AndesGPT
Vivo	Google Gemini, Baidu, DeepSeek	BlueLM
Xiaomi	Google Gemini, Bytedance 豆包, DeepSeek	MiLM-6B

资料来源: BCI

覆盖公司估值表

公司	代码	市值	EPS		P/E		ROE (%)		P/B	
		US mn	2024E	2025E	2024E	2025E	2024E	2025E	2024E	2025E
海外公司-上游										
台积电	2330 TT	904,832	NT\$ 60.7	NT\$ 71.9	17.2x	14.5x	29.2	27.8	5.0x	4.0x
联电	2303 TT	19,460	NT\$ 3.2	NT\$ 3.1	14.5x	15.1x	14.3	14.8	1.5x	1.5x
英伟达	NVDA US	3,480,172	US\$4.38	US\$5.54	32.6x	25.8x	89.9	88.7	29.3x	22.9x
超微半导体	AMD US	197,374	US\$4.05	US\$5.25	30.1x	23.2x	10.8	13.1	3.2x	3.0x
联发科	2454 TT	71,393	NT\$ 72.3	NT\$ 79.9	18.5x	16.7x	29.2	30.3	5.4x	5.1x
比亚迪电子	285 HK	9,172	CNY 2.09	CNY 2.91	15.3x	11.0x	13.6	16.7	2.0x	1.7x
博通	AVGO US	1,148,592	US\$6.47	US\$8.16	37.8x	29.9x	45.1	52.7	16.7x	15.4x
邁威爾	MRVL US	59,613	US\$2.81	US\$3.86	24.6x	17.9x	14.6	16.7	3.6x	3.0x
美光	MU US	123,995	US\$7.32	US\$8.66	15.2x	12.8x	15.8	16.1	2.4x	2.1x
SK海力士	000660 KS	122,734	₩ 43,609	₩ 42,787	5.3x	5.4x	22.9	15.4	1.2x	0.8x
闪迪	SNDK US	6,076	US\$3.04	US\$5.12	13.8x	8.2x	3.8	5.4	0.5x	0.4x
艾司摩尔	ASML US	303,328	EUR24.74	EUR27.69	27.3x	24.4x	47.1	42.5	12.9x	10.4x
陆得斯科技	ONTO US	4,861	US\$5.17	US\$6.34	19.3x	15.7x	12.4	12.5	2.6x	2.4x
ASM PACIFIC	522 HK	3,046	US\$1.43	US\$2.89	40.0x	19.8x	6.5	8.9	1.5x	1.3x
地平线机器人	9660 HK	12,446	(CNY 0.01)	CNY 0.13	-808.8x	62.2x	-0.2	2.9	2.1x	1.7x
英诺赛科	2577 HK	4,152	(CNY 0.94)	(CNY 0.39)	-43.0x	-103.7x	-24.4	-8.3	11.3x	9.2x
海外公司-下游										
苹果	AAPL US	3,008,822	US\$7.19	US\$7.69	28.0x	26.2x	237.5	487.5	90.2x	237.1x
戴尔	DELL US	78,067	US\$9.31	US\$10.26	12.3x	11.1x	N/A	N/A	N/A	N/A
联想	992 HK	14,540	HK\$0.99	HK\$1.29	9.3x	7.1x	24.8	26.7	16.9x	13.8x
小米集团	1810 HK	176,715	CNY 1.76	CNY 2.20	30.3x	24.3x	19.0	19.2	5.8x	4.7x
鸿海	2317 TT	72,120	NT\$ 13.2	NT\$ 15.1	11.8x	10.3x	10.5	11.3	1.2x	1.2x
台光电	2383 TT	10,305	US\$41.98	US\$50.87	21.2x	17.5x	33.6	24.2	7.1x	4.2x
建滔积层板	1888 HK	3,971	US\$0.66	US\$0.78	15.2x	12.8x	18.0	18.6	1.6x	1.4x
天弘科技	CLS US	14,084	US\$5.38	US\$6.64	22.8x	18.5x	25.5	20.4	5.8x	3.7x
艾瑞斯塔	ANET US	121,573	US\$2.75	US\$3.18	35.1x	30.5x	24.9	20.9	8.7x	6.4x
Coherent	COHR US	12,620	US\$3.66	US\$4.39	22.2x	18.5x	15.4	8.5	1.9x	1.6x
Lumentum	LITE US	5,698	US\$1.86	US\$4.04	44.2x	20.3x	25.3	25.8	11.2x	5.2x
Fabrinet	FN US	8,342	US\$10.08	US\$11.05	23.1x	21.1x	24.4	36.5	6.3x	7.7x

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